

University of California Riverside
Electrical and Computer Engineering Department

EE 116, Electromagnetics I

Lecture 2: Vector Algebra

Chapter 1: 1.1-1.7

04/02/2026

Instructor: Dr. Alexander Khitun

Email: akhitun@ece.ucr.edu

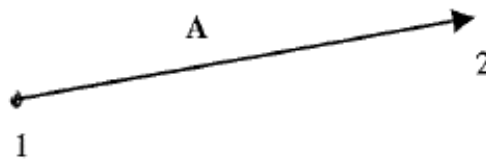
Lecture Outline

- Definitions
- Summation and Multiplication of Vectors
- Systems of coordinates
- Practice Session
- Vector Field
- Line and Surface Integrals

Vector Algebra

- Scalar: is a physical quantity completely specified by a single number describing its magnitude.
 - *Example: temperature*
- Vector: is a physical quantity completely specified by both its magnitude and direction.
 - *Example: velocity of a car*

Vector representation by an arrow

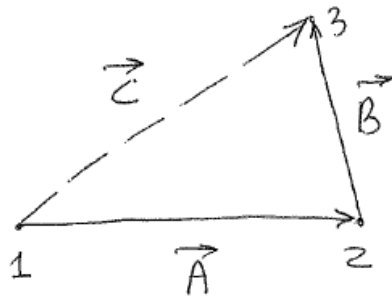


- The length of the arrow is proportional to the magnitude of the vector
- Direction of the vector is indicated by the direction of the arrow

$$\mathbf{A} = |\mathbf{A}| \mathbf{a}_x, \quad \mathbf{a}_x \text{ is a unit vector}$$

Vector Addition and Subtraction

$$\vec{C} = \vec{A} + \vec{B}$$

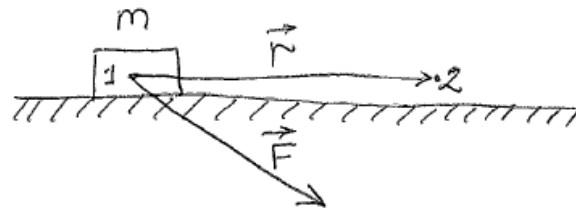


$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

$$\begin{aligned} (\vec{A} + \vec{B}) + \vec{D} &= \\ &= \vec{A} + (\vec{B} + \vec{D}) \end{aligned}$$

Scalar (Dot) Product of Two Vectors

$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \alpha$, where α is the angle between $\mathbf{A}$ and $\mathbf{B}$.



The work done to move the object:

$$W = |\vec{F}| \cdot \cos \alpha \cdot |\vec{r}|$$

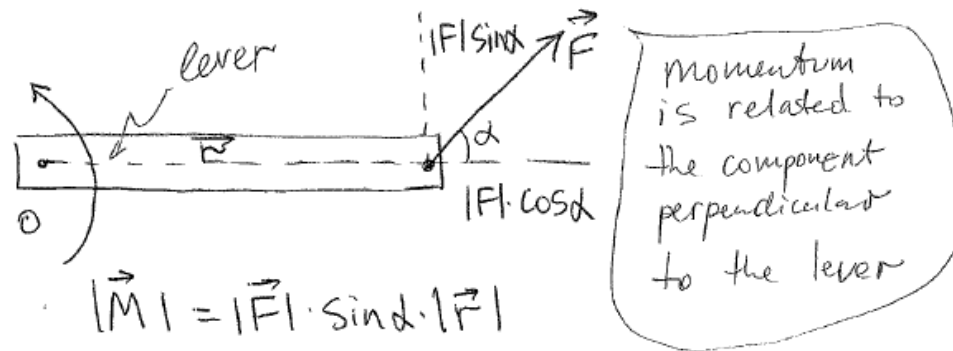
$$W = \vec{F} \cdot \vec{r}$$

If $\vec{a} \perp \vec{b}$, what is $\vec{a} \cdot \vec{b}$?

Vector (Cross) Product of Two Vectors

$$\vec{C} = \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \alpha \vec{a}_c, \text{ where } \alpha \text{ is the angle between } \vec{A} \text{ and } \vec{B}.$$

The direction of $\vec{C}$ is obtained by the "right-hand rule".



-magnitude of the momentum

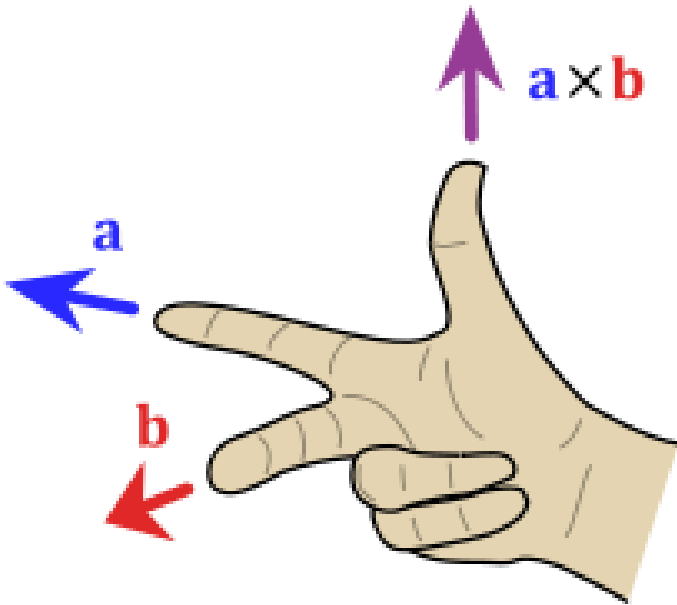
$$\vec{M} = \vec{r} \times \vec{F}$$

$$\vec{B} \times \vec{A} = -\vec{C} = -\vec{A} \times \vec{B}$$

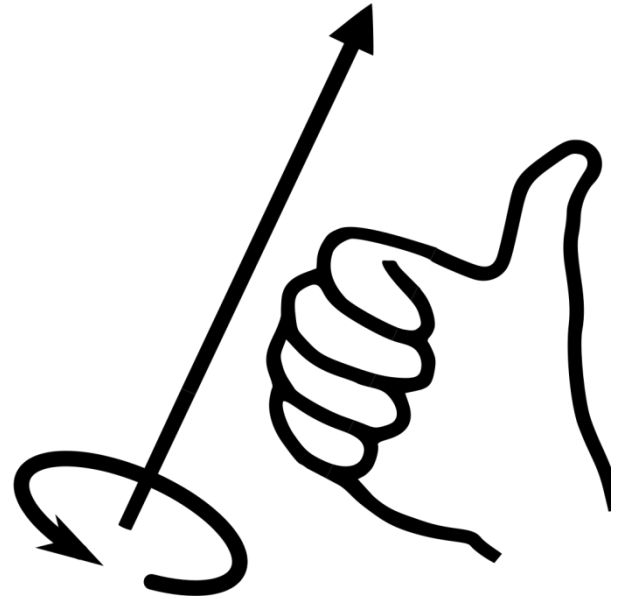
$\Rightarrow$ cross product does not obey a commutative law

If $\vec{a} \parallel \vec{b}$ what is $\vec{a} \times \vec{b}$?

Right-hand Rule



Finding the direction of the cross product by the right-hand rule



Right hand rule for curve orientation.

Basis and Unit Vectors

A unit vector might be a basis vector, and vice-versa: a unit vector is simply a vector whose magnitude is 1, while a basis vector is an element of a basis of a vector space V , that is, a set of vectors that span (i.e., generate by means of their linear combinations) the vector space V .

A **basis** is a set of vectors that:

- **Span the space** (you can build any vector from them)
- **Are linearly independent**

👉 Example (Cartesian basis):

$$\{\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z\}$$

Any vector can be written as:

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$$

A **unit vector** is any vector with **magnitude = 1**:

$$|\mathbf{u}| = 1$$

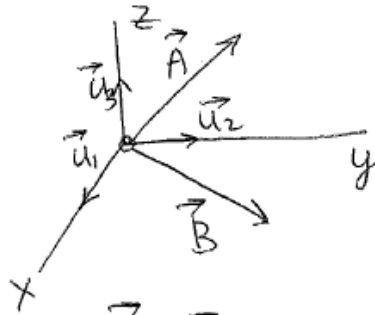
$$\mathbf{u} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

- It represents **direction only**
- It does not need to be part of a basis

VECTOR REPRESENTATION IN THE COORDINATE SYSTEMS

The vector $\vec{A}$ can be represented in terms of its components

$$\vec{A} = A_x \cdot \vec{a}_x + A_y \vec{a}_y + A_z \vec{a}_z$$



$$\vec{A} = A_1 \vec{u}_1 + A_2 \vec{u}_2 + A_3 \vec{u}_3$$

$$\vec{B} = B_1 \vec{u}_1 + B_2 \vec{u}_2 + B_3 \vec{u}_3$$

$$\vec{A} + \vec{B} = (A_1 + B_1) \vec{u}_1 + (A_2 + B_2) \vec{u}_2 + (A_3 + B_3) \vec{u}_3$$

Properties of unit vectors:

$$\vec{u}_1 \cdot \vec{u}_2 = 0$$

$$\vec{u}_1 \cdot \vec{u}_1 = 1$$

$$\vec{A} \cdot \vec{B} = (A_1 \vec{u}_1 + A_2 \vec{u}_2 + A_3 \vec{u}_3) \cdot (B_1 \vec{u}_1 + B_2 \vec{u}_2 + B_3 \vec{u}_3) =$$

$$= A_1 B_1 + A_2 B_2 + A_3 B_3$$

The cross product

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} =$$

$$= \vec{u}_1 \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \vec{u}_2 \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \vec{u}_3 \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix} =$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) - \vec{u}_2 (A_1 B_3 - A_3 B_1) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) + \vec{u}_2 (A_3 B_1 - A_1 B_3) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

This follows from the properties

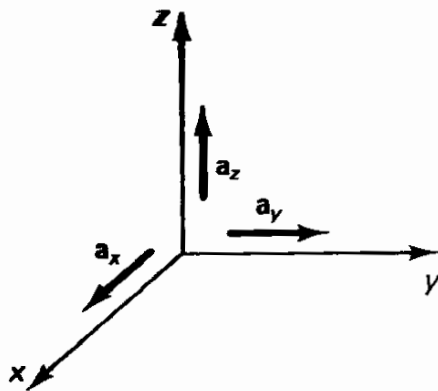
$$\vec{u}_1 \times \vec{u}_2 = \vec{u}_3, \quad \vec{u}_2 \times \vec{u}_3 = \vec{u}_1, \quad \vec{u}_3 \times \vec{u}_1 = \vec{u}_2$$

$$\vec{u}_1 \times \vec{u}_1 = \vec{u}_2 \times \vec{u}_2 = \vec{u}_3 \times \vec{u}_3 = \vec{0}$$

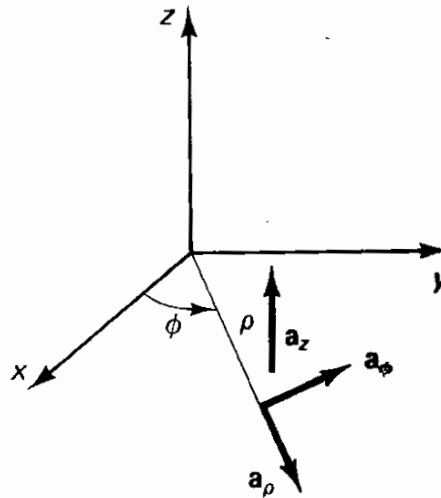
Coordinate Systems

In electromagnetics, the fields are functions of space and time. A three-dimensional coordinate system allow us to uniquely specify the location of a point in space or the direction of a vector quantity.

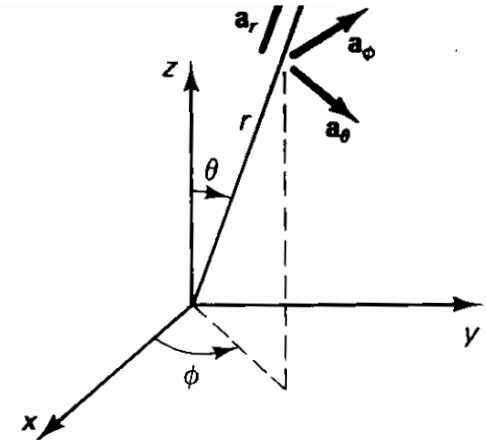
- Cartesian (rectangular) coordinate system
- Cylindrical coordinate system
- Spherical



Cartesian



Cylindrical



Spherical

Figure 1.18 The base vectors of the three most commonly used coordinate systems.

Cartesian Coordinates

(x,y,z)

Differential length:

$$d\mathbf{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz$$

Differential surface area:

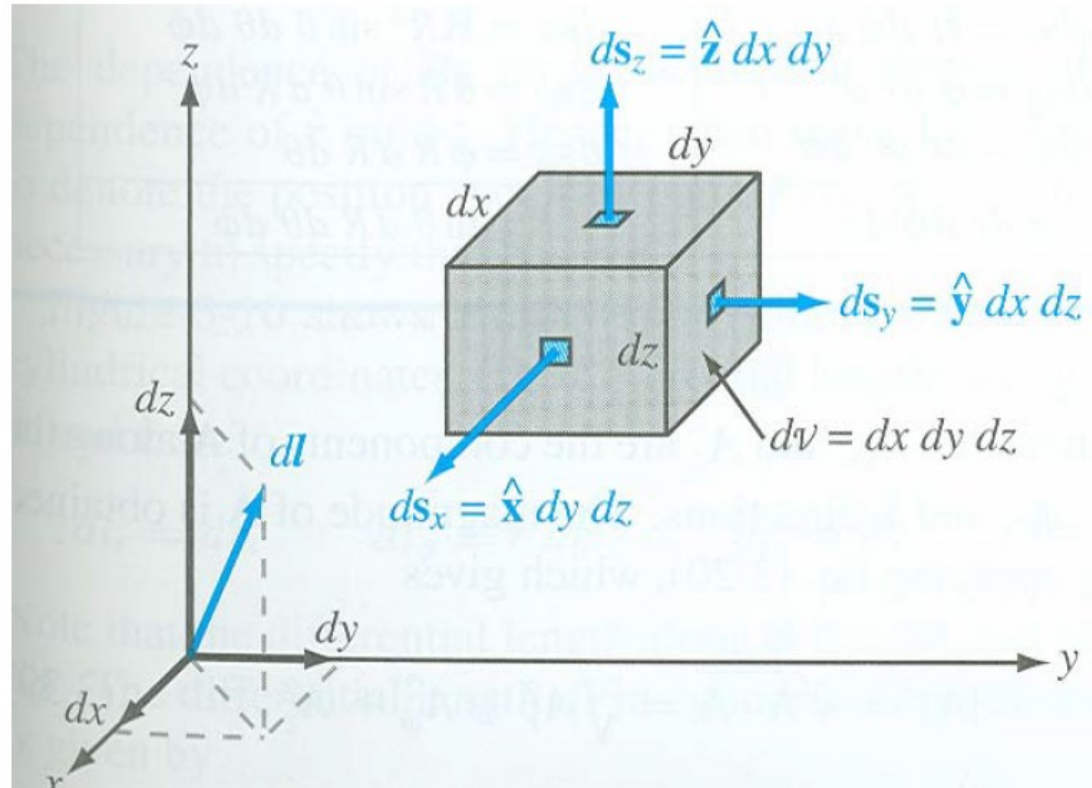
$$d\mathbf{s}_x = \hat{x}dydz$$

$$d\mathbf{s}_y = \hat{y}dxdz$$

$$d\mathbf{s}_z = \hat{z}dxdy$$

Differential volume:

$$dv = dxdydz$$



Point: (x,y,z)

$$x, y, z \in \{-\infty, +\infty\}$$

Vector: $= A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$

Right-hand Rule in Cartesian Coordinate System

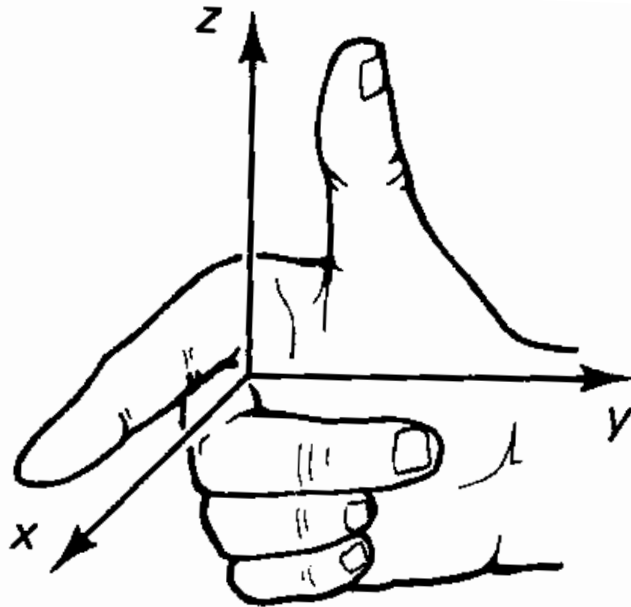
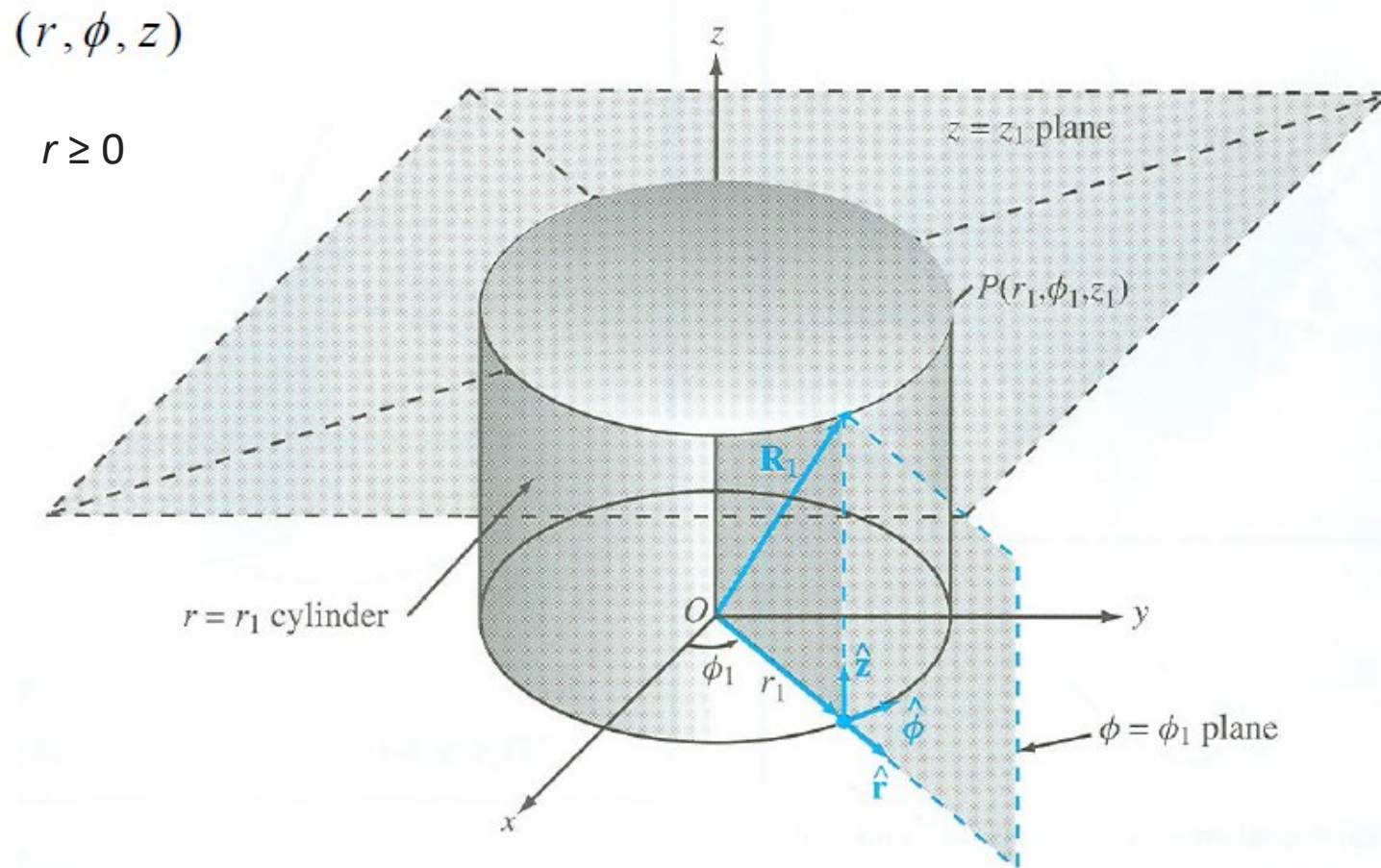


Figure 1.12 The coordinate axes in the Cartesian coordinate system are mutually orthogonal and the rotation from two of them toward the third axis follows the right-hand rule.

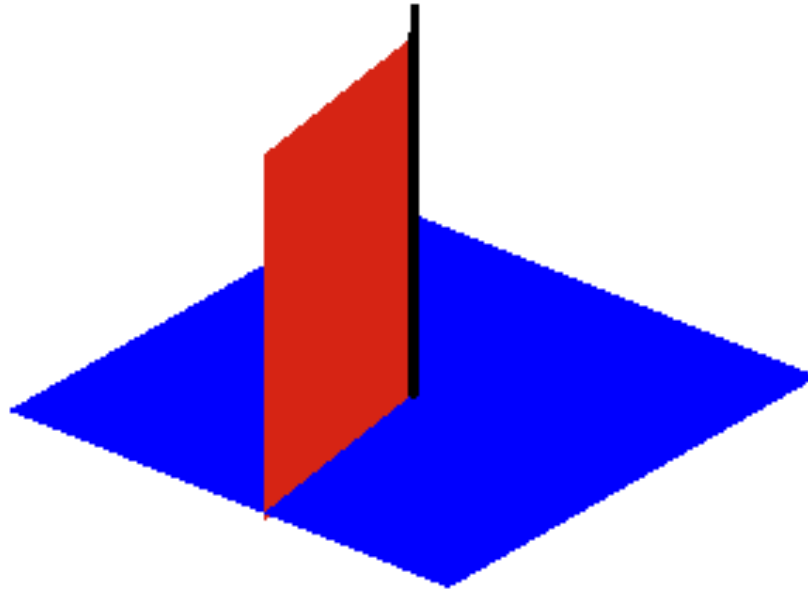
Cylindrical Coordinates



Point: (ρ, ϕ, z)

Vector: $= A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$

Cylindrical Coordinates



Cylindrical Coordinates

Differential length:

$$d\mathbf{l} = \hat{r}dr + \hat{\phi}r d\phi + \hat{z}dz$$

Differential surface area:

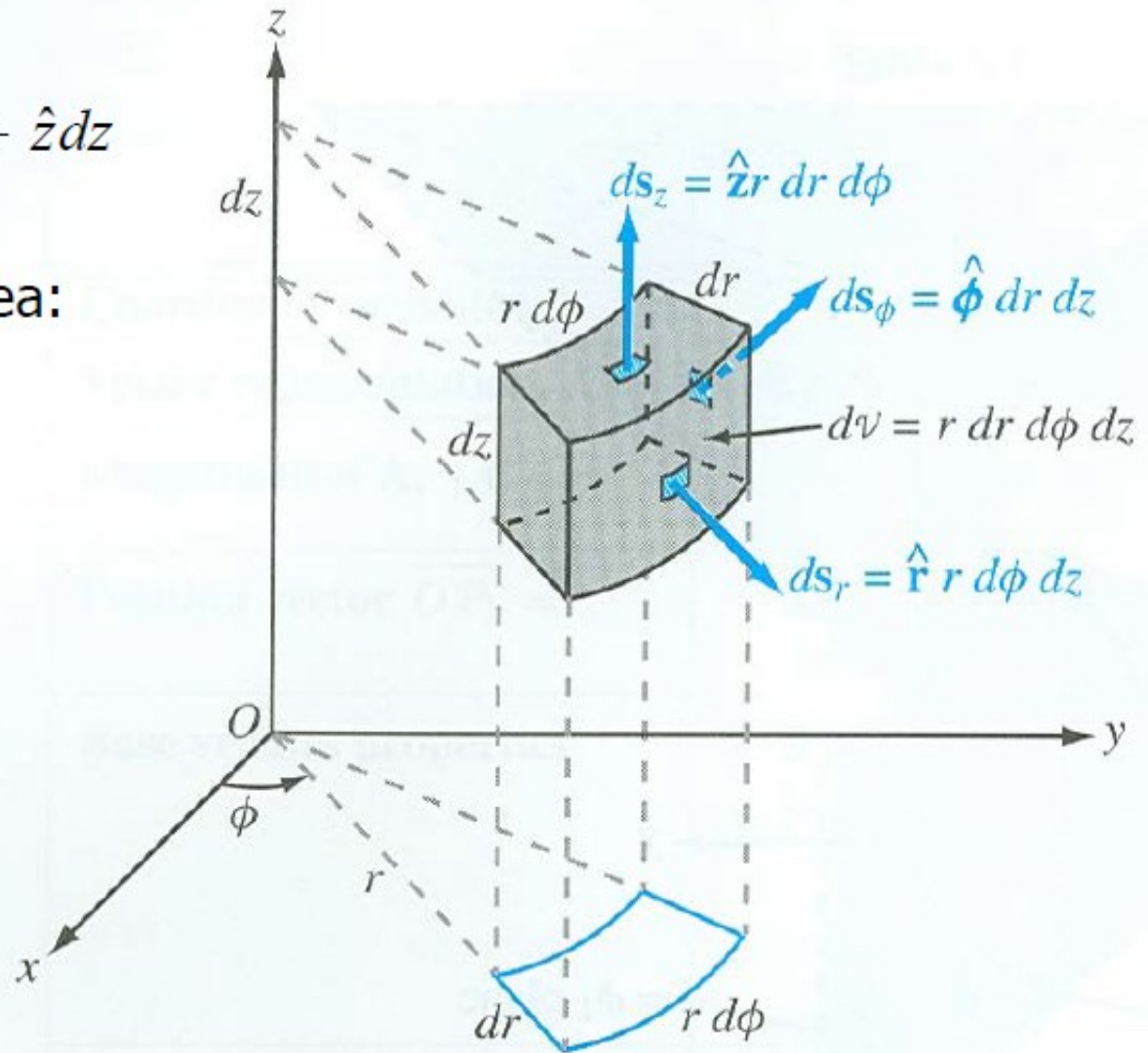
$$ds_r = \hat{r}r d\phi dz$$

$$ds_\phi = \hat{\phi}dr dz$$

$$ds_z = \hat{z}r d\phi dr$$

Differential volume:

$$dv = r dr d\phi dz$$



Cylindrical Coordinate System

$$dv = d\rho(\rho d\phi)dz = \rho d\rho d\phi dz$$

The resultant element of length $d\ell$ from P_1 to P_2 is given by

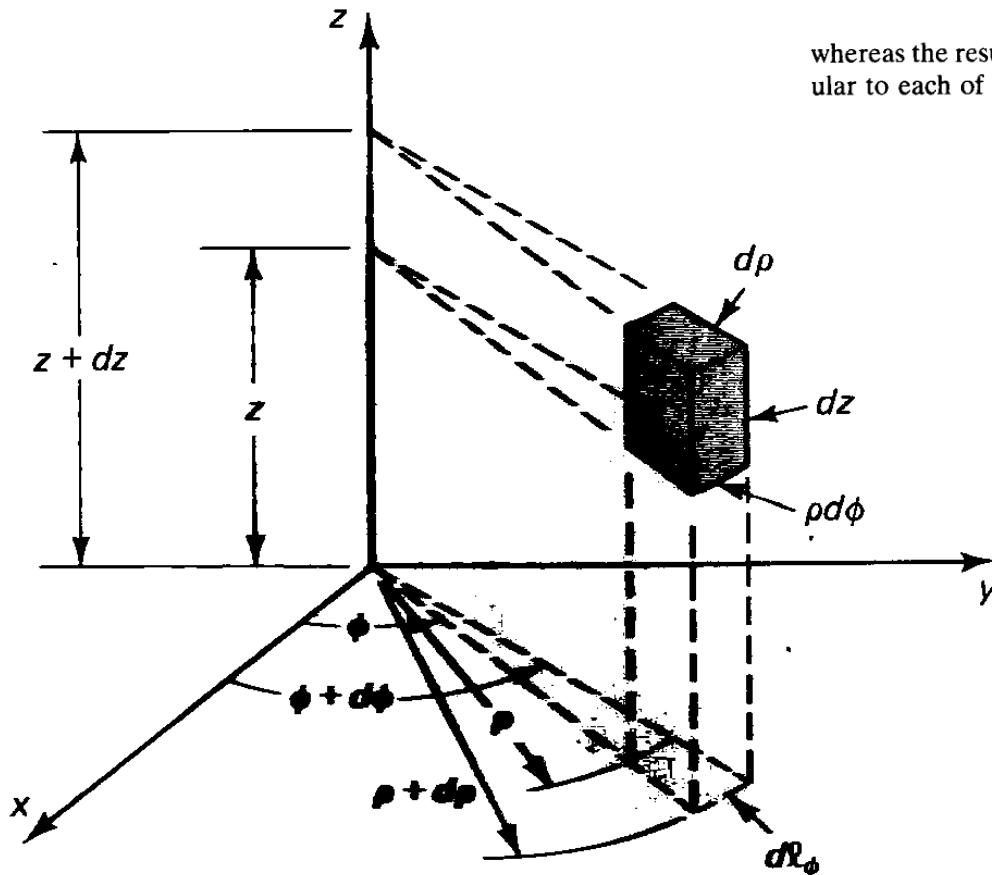
$$d\ell = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$

whereas the resultant elements of area **that are** associated with unit vectors perpendicular to each of the areas to emphasize **their** orientations are given by

$$ds_\rho = \rho d\phi dz \mathbf{a}_\rho$$

$$ds_\phi = d\rho dz \mathbf{a}_\phi$$

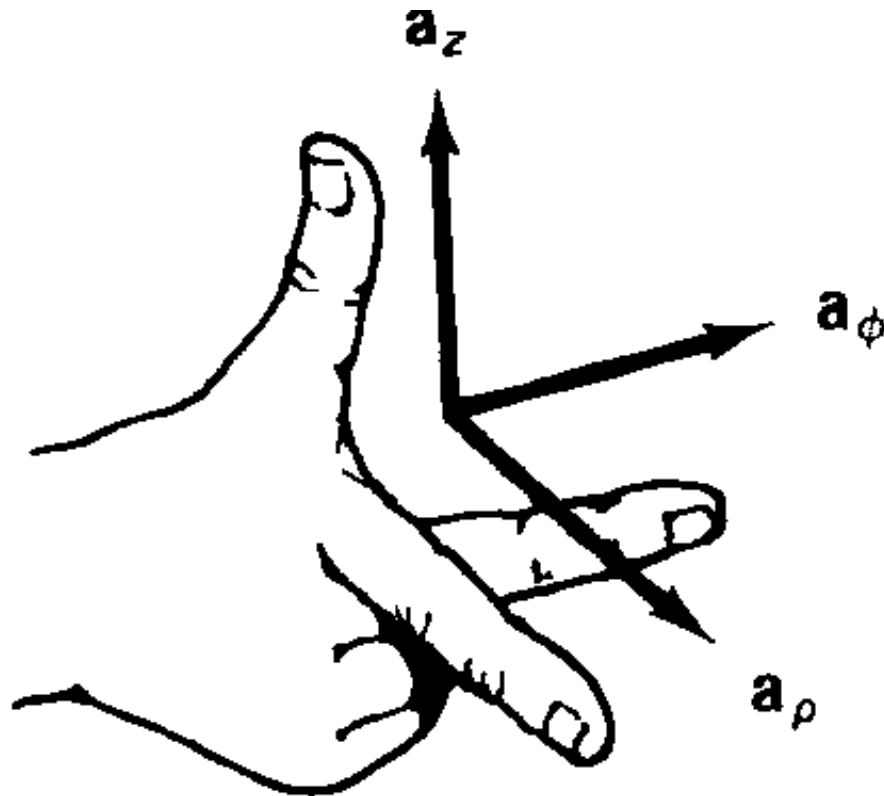
$$ds_z = \rho d\rho d\phi \mathbf{a}_z$$



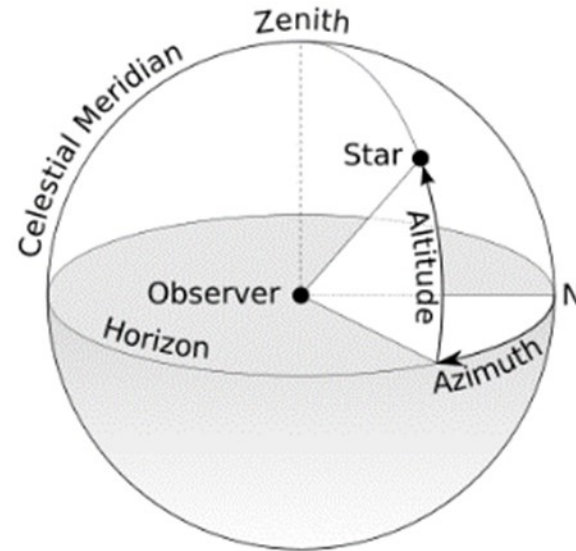
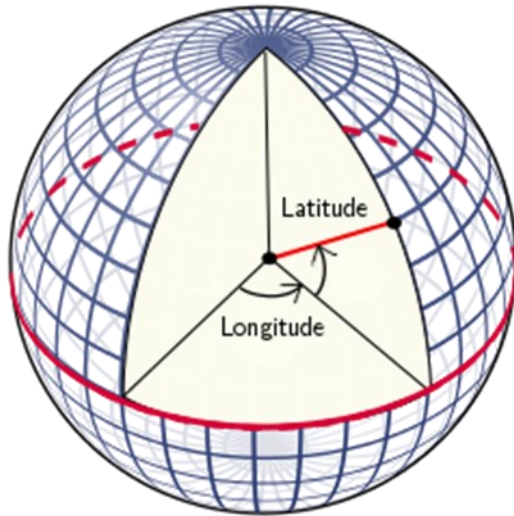
$$\begin{aligned} d\ell &= d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z \\ ds_\rho &= \rho d\phi dz \mathbf{a}_\rho, \quad ds_\phi = d\rho dz \mathbf{a}_\phi \\ ds_z &= \rho d\rho d\phi \mathbf{a}_z \\ dv &= \rho d\rho d\phi dz \end{aligned}$$

Figure 1.15 The elements of length, volume, and surface in the cylindrical coordinate system. $d\phi$ is not a length and should be multiplied by its metric coefficient ρ to have the dimension of length.
 $\therefore d\ell_\phi = \rho d\phi \mathbf{a}_\phi$.

Right-hand Rule in Cylindrical Coordinate System



Spherical Coordinate System



An azimuthal angle is the angular measurement of a point's position in a horizontal plane, typically measured clockwise in degrees from a reference direction like North. It defines direction, such as identifying where a satellite is in the sky or how a solar panel is oriented.

The polar angle is called "polar" because it measures the orientation of a point relative to a fixed, central point called the pole (analogous to the origin in Cartesian coordinates) and a fixed polar axis. It defines the angle of rotation around this central pole, serving as the angular coordinate in a polar coordinate system.

Spherical Coordinates

Differential length:

$$d\mathbf{l} = \hat{R}dR + \hat{\theta}Rd\theta + \hat{\phi}R \sin \theta d\phi$$

Differential surface area:

$$ds_R = \hat{R}R^2 \sin \theta d\theta d\phi$$

$$ds_\theta = \hat{\theta}R \sin \theta dR d\phi$$

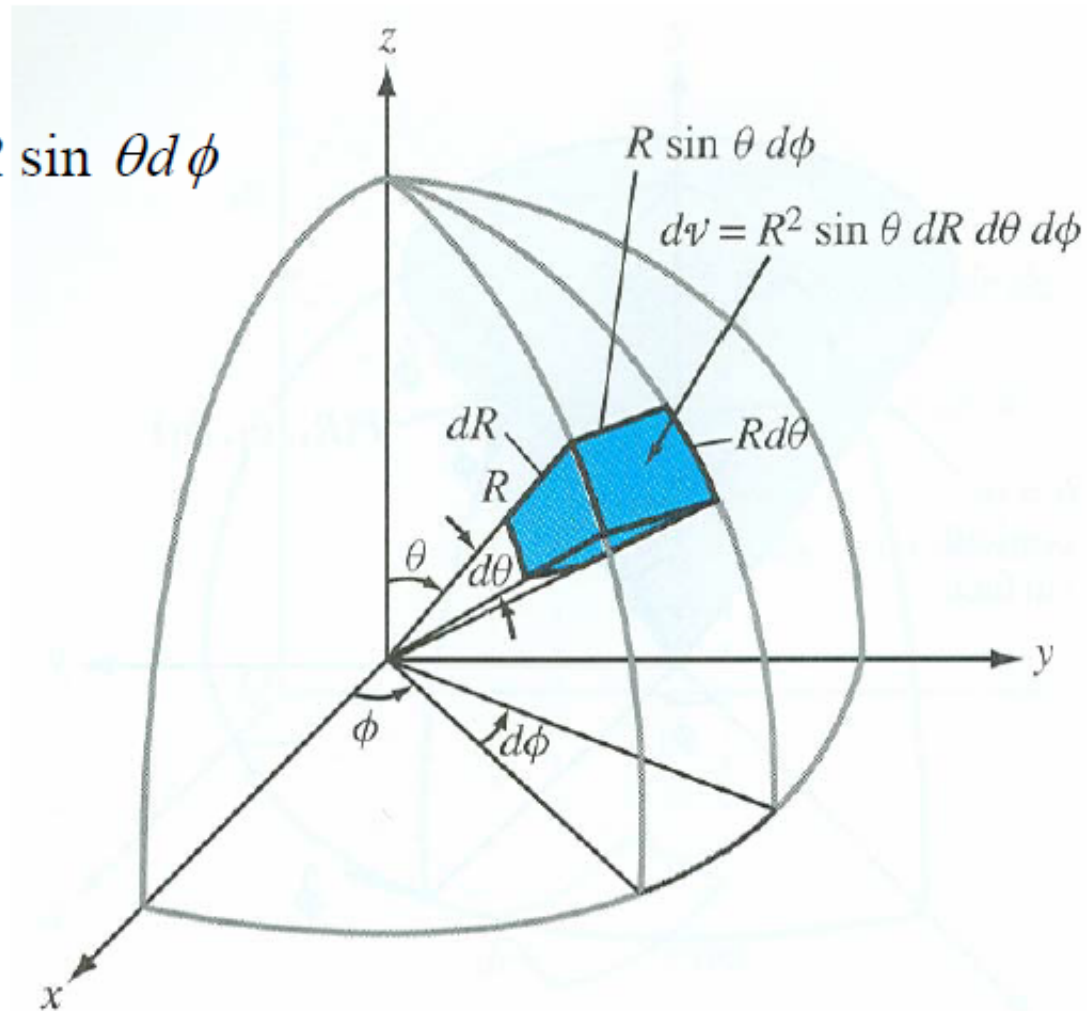
$$ds_\phi = \hat{\phi}RdRd\theta$$

Differential volume:

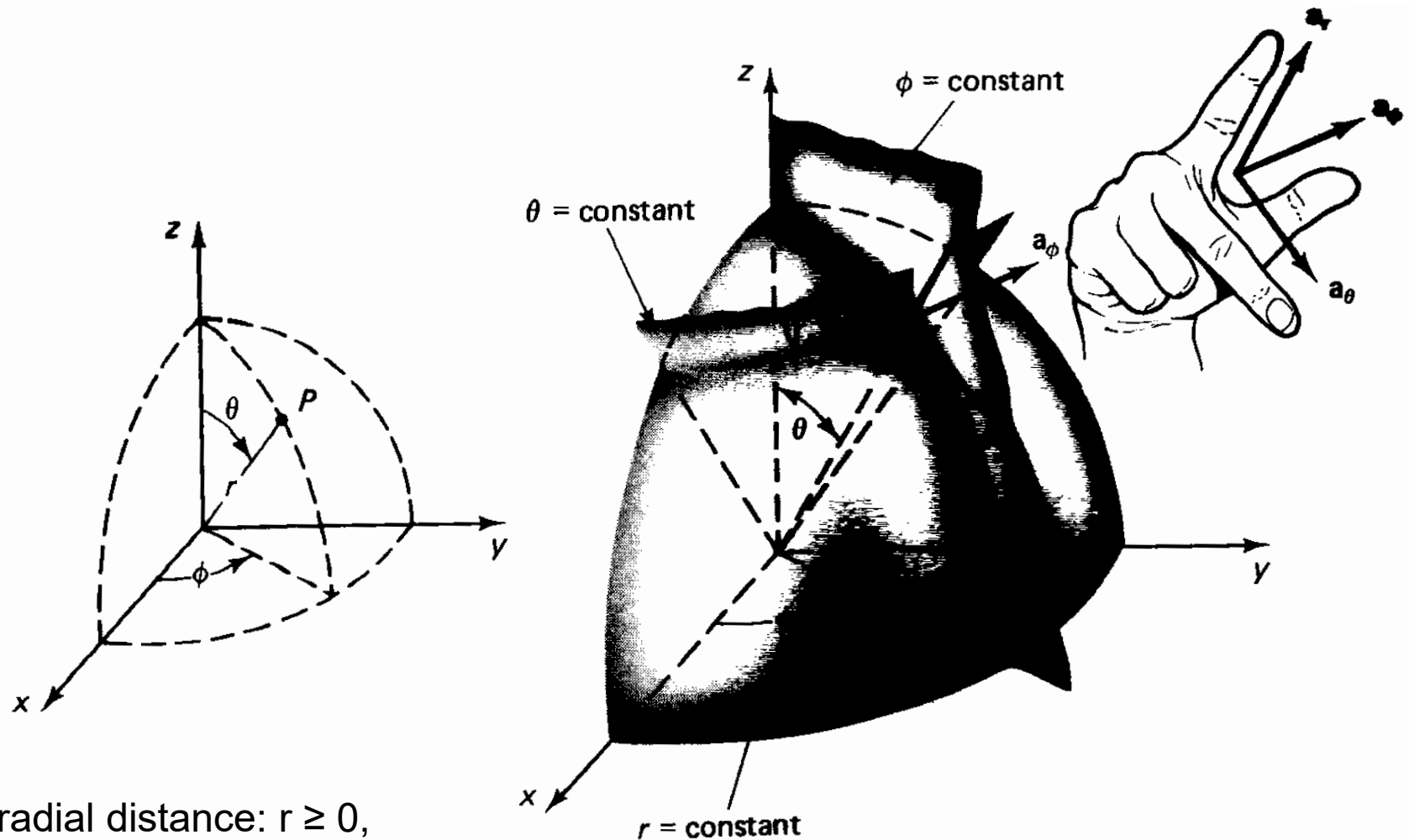
$$dv = R^2 \sin \theta dR d\theta d\phi$$

Point: (r, θ, ϕ)

Vector: $= A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$

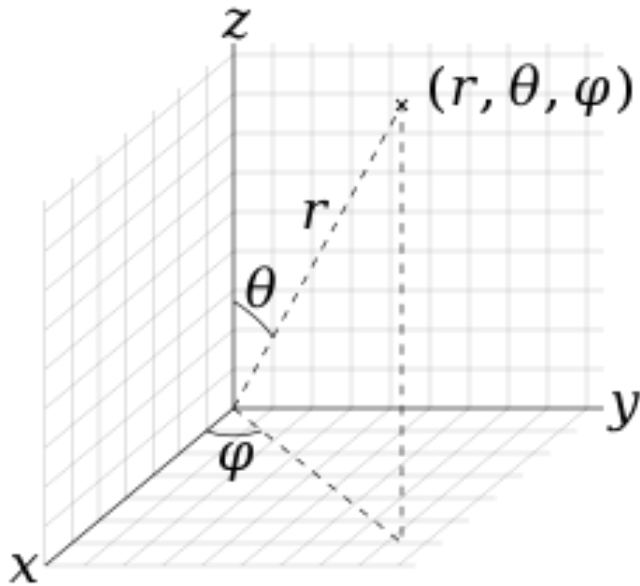


Spherical Coordinate System

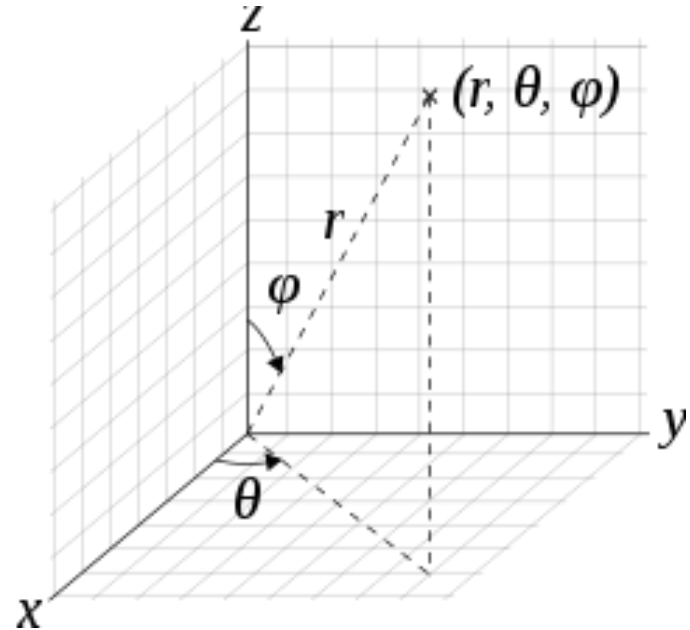


radial distance: $r \geq 0$,
polar angle: $0^\circ \leq \theta \leq 180^\circ$ (π rad),
azimuth : $0^\circ \leq \phi < 360^\circ$ (2π rad).

Attention!



Spherical coordinates (r, θ, φ) as commonly used in **physics** ([ISO 80000-2:2019](#) convention): radial distance r ([slant distance](#) to origin), polar angle θ ([theta](#)) (angle with respect to positive polar axis), and azimuthal angle φ ([phi](#)) (angle of rotation from the initial meridian plane).



Spherical coordinates (r, θ, φ) as often used in **mathematics**: radial distance r , azimuthal angle θ , and polar angle φ . The meanings of θ and φ have been swapped compared to the **physics** convention. As in physics, ρ ([rho](#)) is often used instead of r , to avoid confusion with the value r in cylindrical and 2D polar coordinates.

Base Vectors

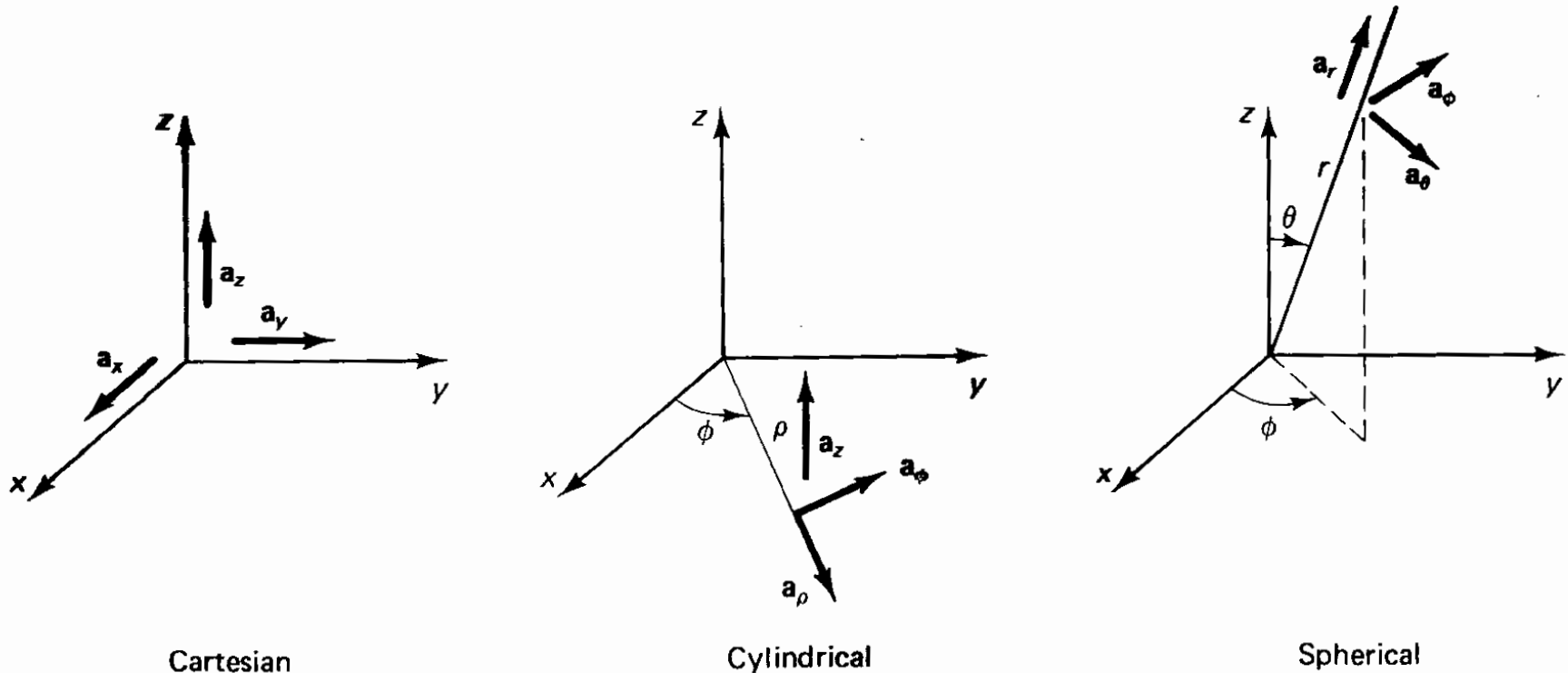


Figure 1.18 The base vectors of the three most **commonly** used coordinate systems.

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z \text{ (Cartesian system)}$$

$$\mathbf{A} = A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z \text{ (Cylindrical system)}$$

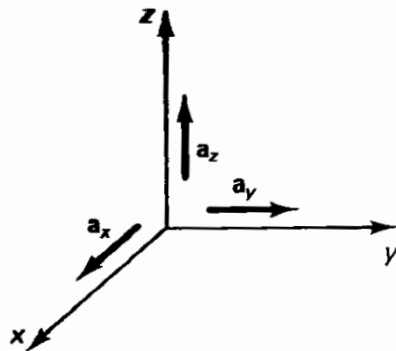
$$\mathbf{A} = A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi \text{ (Spherical system)}$$

Practice Session

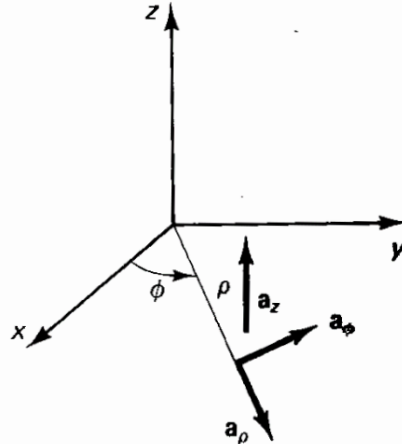
EXAMPLE 1.1

Which of the following sets of independent variables (coordinates) define a point in a coordinate system?

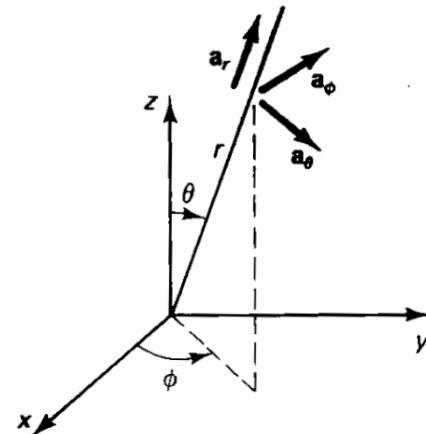
1. $x = 2, y = -4, z = 0$.
2. $\rho = -4, \phi = 0^\circ, z = -1$.
3. $r = 3, \theta = -90^\circ, \phi = 0^\circ$.



Cartesian



Cylindrical



Spherical

Figure 1.18 The base vectors of the three most **commonly** used coordinate systems.

Practice Session

- Given two vectors A and B:

$$\mathbf{A} = 2\mathbf{a}_x - 1\mathbf{a}_z$$

$$\mathbf{B} = 3\mathbf{a}_y$$

Calculate:

a) $\mathbf{A} + \mathbf{B}$

$$\mathbf{A} + \mathbf{B} = 2\mathbf{a}_x + 3\mathbf{a}_y - 1\mathbf{a}_z$$

b) $\mathbf{A} \cdot \mathbf{B}$

$$\mathbf{A} \cdot \mathbf{B} = 0$$

c) $\mathbf{A} \times \mathbf{B}$

$$\mathbf{A} \times \mathbf{B} = 3\mathbf{a}_x + 6\mathbf{a}_z$$

d) $\mathbf{B} \times \mathbf{A}$

$$\mathbf{B} \times \mathbf{A} = -3\mathbf{a}_x - 6\mathbf{a}_z$$

Practice Session

- Given two vectors A and B:

$$\mathbf{A} = 5\mathbf{a}_x - 4\mathbf{a}_y + 2\mathbf{a}_z$$

$$\mathbf{B} = -1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z$$

Calculate:

a) $\mathbf{A} + \mathbf{B}$

$$\mathbf{A} + \mathbf{B} = 4\mathbf{a}_x - 2\mathbf{a}_y + 9\mathbf{a}_z$$

b) $\mathbf{A} \cdot \mathbf{B}$

$$\mathbf{A} \cdot \mathbf{B} = -5 - 8 + 14 = 1$$

c) $\mathbf{A} \times \mathbf{B}$

d) $\mathbf{B} \times \mathbf{A}$

Practice Session

- Given two vectors A and B

$$\mathbf{A} = 5\mathbf{a}_x - 4\mathbf{a}_y + 2\mathbf{a}_z$$

$$\mathbf{B} = -1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z$$

Calculate:

a) $\mathbf{A} + \mathbf{B}$

b) $\mathbf{A} \cdot \mathbf{B}$

c) $\mathbf{A} \times \mathbf{B}$

d) $\mathbf{B} \times \mathbf{A}$

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= [(-4 \cdot 7) - (2 \cdot 2)]\mathbf{a}_x + \\ &[(2 \cdot -1) - (5 \cdot 7)]\mathbf{a}_y + \\ &[(5 \cdot 2) - (-4 \cdot -1)]\mathbf{a}_z = \\ &-32\mathbf{a}_x - 37\mathbf{a}_y + 6\mathbf{a}_z \end{aligned}$$

The cross product

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{u}_1 & \vec{u}_2 & \vec{u}_3 \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix} =$$

$$= \vec{u}_1 \begin{vmatrix} A_2 & A_3 \\ B_2 & B_3 \end{vmatrix} - \vec{u}_2 \begin{vmatrix} A_1 & A_3 \\ B_1 & B_3 \end{vmatrix} + \vec{u}_3 \begin{vmatrix} A_1 & A_2 \\ B_1 & B_2 \end{vmatrix} =$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) - \vec{u}_2 (A_1 B_3 - A_3 B_1) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

$$= \vec{u}_1 (A_2 B_3 - A_3 B_2) + \vec{u}_2 (A_3 B_1 - A_1 B_3) + \vec{u}_3 (A_1 B_2 - A_2 B_1)$$

This follows from the properties

$$\vec{u}_1 \times \vec{u}_2 = \vec{u}_3, \quad \vec{u}_2 \times \vec{u}_3 = \vec{u}_1, \quad \vec{u}_3 \times \vec{u}_1 = \vec{u}_2$$

$$\vec{u}_1 \times \vec{u}_1 = \vec{u}_2 \times \vec{u}_2 = \vec{u}_3 \times \vec{u}_3 = 0$$

- Given two vectors A and B:

$$\mathbf{A} = 5\mathbf{a}_x - 4\mathbf{a}_y + 2\mathbf{a}_z$$

$$\mathbf{B} = -1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z$$

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= 5\mathbf{a}_x \times [-1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z] + -4\mathbf{a}_y \times [-1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z] + \\ &2\mathbf{a}_z [-1\mathbf{a}_x + 2\mathbf{a}_y + 7\mathbf{a}_z] = 0 + 10\mathbf{a}_z - 35\mathbf{a}_y - 4\mathbf{a}_z + 0 - 28\mathbf{a}_x \\ &- 2\mathbf{a}_y - 4\mathbf{a}_x + 0 = -32\mathbf{a}_x - 37\mathbf{a}_y + 6\mathbf{a}_z\end{aligned}$$

Practice Session Cylindrical Coordinates

$$\mathbf{A} = 3 \mathbf{a}_\rho + 2 \mathbf{a}_\phi$$

$$\mathbf{B} = 1 \mathbf{a}_\rho + 4 \mathbf{a}_\phi$$

$$\mathbf{A} + \mathbf{B} = (3 + 1)\mathbf{a}_\rho + (2 + 4)\mathbf{a}_\phi$$

$$\boxed{\mathbf{A} + \mathbf{B} = 4 \mathbf{a}_\rho + 6 \mathbf{a}_\phi}$$

Calculating the dot and cross products

$$\mathbf{A} = 3 \mathbf{a}_\rho + 2 \mathbf{a}_\phi$$

$$\mathbf{B} = 1 \mathbf{a}_\rho + 4 \mathbf{a}_\phi$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = 1, \quad \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1, \quad \mathbf{a}_\rho \cdot \mathbf{a}_\phi = 0$$

$$\mathbf{A} \cdot \mathbf{B} = 3 \cdot 1 + 2 \cdot 4 = 3 + 8 = 11$$

$$\boxed{\mathbf{A} \cdot \mathbf{B} = 11}$$

Calculating the cross products

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{A} \times \mathbf{B} = (3\mathbf{a}_\rho + 2\mathbf{a}_\phi) \times (1\mathbf{a}_\rho + 4\mathbf{a}_\phi)$$

$$= 3 \cdot 4(\mathbf{a}_\rho \times \mathbf{a}_\phi) + 2 \cdot 1(\mathbf{a}_\phi \times \mathbf{a}_\rho)$$

$$= 12\mathbf{a}_z - 2\mathbf{a}_z = 10\mathbf{a}_z$$

Practice Example: Vector Operations in Spherical Coordinates

$$\mathbf{A} = 2 \mathbf{a}_r + 3 \mathbf{a}_\theta + 1 \mathbf{a}_\phi$$

$$\mathbf{B} = 1 \mathbf{a}_r + 2 \mathbf{a}_\theta + 4 \mathbf{a}_\phi$$

- Orthonormal:

$$\mathbf{a}_r \cdot \mathbf{a}_r = 1, \quad \mathbf{a}_\theta \cdot \mathbf{a}_\theta = 1, \quad \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

- Cross terms = 0
- Cross products:

$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi, \quad \mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r, \quad \mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$

1. Vector Addition

$$\mathbf{A} + \mathbf{B} = (2 + 1)\mathbf{a}_r + (3 + 2)\mathbf{a}_\theta + (1 + 4)\mathbf{a}_\phi$$

$$\boxed{\mathbf{A} + \mathbf{B} = 3 \mathbf{a}_r + 5 \mathbf{a}_\theta + 5 \mathbf{a}_\phi}$$

$$\mathbf{A} = 2 \mathbf{a}_r + 3 \mathbf{a}_\theta + 1 \mathbf{a}_\phi$$

$$\mathbf{B} = 1 \mathbf{a}_r + 2 \mathbf{a}_\theta + 4 \mathbf{a}_\phi$$

2. Dot Product

$$\mathbf{A} \cdot \mathbf{B} = 2 \cdot 1 + 3 \cdot 2 + 1 \cdot 4 = 2 + 6 + 4$$

$$\boxed{\mathbf{A} \cdot \mathbf{B} = 12}$$

3. Cross Product (Step-by-step)

$$\mathbf{A} \times \mathbf{B} = (2\mathbf{a}_r + 3\mathbf{a}_\theta + 1\mathbf{a}_\phi) \times (1\mathbf{a}_r + 2\mathbf{a}_\theta + 4\mathbf{a}_\phi)$$

Expand using determinant-like pattern:

$$\begin{aligned} & (3 \cdot 4 - 1 \cdot 2)\mathbf{a}_r \\ & = +(1 \cdot 1 - 2 \cdot 4)\mathbf{a}_\theta \\ & \quad +(2 \cdot 2 - 3 \cdot 1)\mathbf{a}_\phi \end{aligned}$$

$$= (12 - 2)\mathbf{a}_r + (1 - 8)\mathbf{a}_\theta + (4 - 3)\mathbf{a}_\phi$$

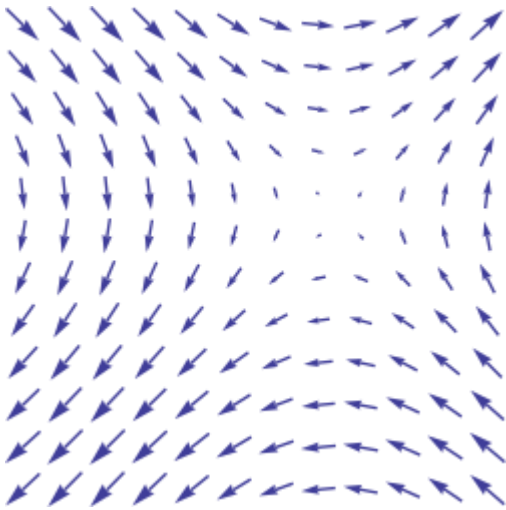
$$\boxed{\mathbf{A} \times \mathbf{B} = 10 \mathbf{a}_r - 7 \mathbf{a}_\theta + 1 \mathbf{a}_\phi}$$

Table 2-2: Coordinate transformation relations.

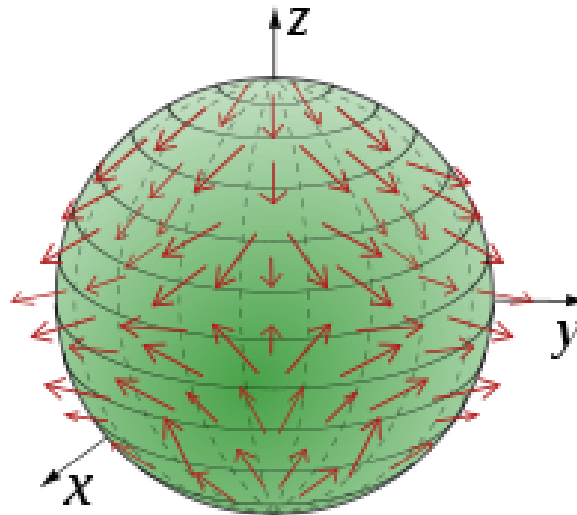
Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$ $\hat{z} = \hat{z}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{x} = \hat{r} \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{r} \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{z}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{R} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta$ $\hat{\theta} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta$ $\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{x} = \hat{R} \sin \theta \cos \phi + \hat{\theta} \cos \theta \cos \phi - \hat{\phi} \sin \phi$ $\hat{y} = \hat{R} \sin \theta \sin \phi + \hat{\theta} \cos \theta \sin \phi + \hat{\phi} \cos \phi$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{R} = \hat{r} \sin \theta + \hat{z} \cos \theta$ $\hat{\theta} = \hat{r} \cos \theta - \hat{z} \sin \theta$ $\hat{\phi} = \hat{\phi}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{r} = \hat{R} \sin \theta + \hat{\theta} \cos \theta$ $\hat{\phi} = \hat{\phi}$ $\hat{z} = \hat{R} \cos \theta - \hat{\theta} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

Vector Field

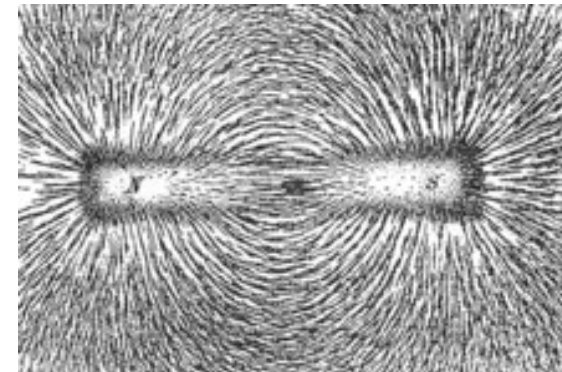
- In vector calculus and physics, a vector field is an assignment of a vector to each point in a subset of space.



A portion of the vector field
($\sin y, \sin x$)



A vector field on a sphere



Magnetic field lines of
an iron bar (magnetic
dipole)

Flux

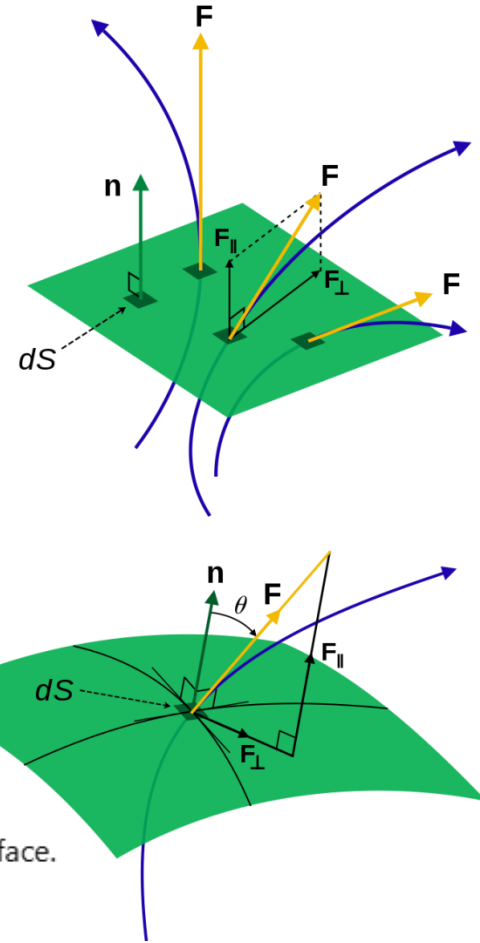
- The word flux comes from Latin: fluxus means "flow", and fluere is "to flow". As fluxion, this term was introduced into differential calculus by Isaac Newton. Flux describes any effect that appears to pass or travel (whether it actually moves or not) through a surface or substance. A flux is either a concept based in physics or used with applied mathematics.
- In the case of fluxes, we have to take the integral, over a surface, of the flux through every element of the surface. The result of this operation is called the surface integral of the flux. It represents the quantity which passes through the surface.
- — James Clerk Maxwell

Key Concept

A vector field can be interpreted as a **flux density** — the amount of quantity passing through a surface.

$$\Phi = \mathbf{A} \cdot \mathbf{S}$$

- $\mathbf{A}$: vector field (e.g., velocity, electric field)
- $\mathbf{S} = S \mathbf{a}_n$: surface vector (area $\times$ normal)
- Φ : flux through the surface



Surface Integral

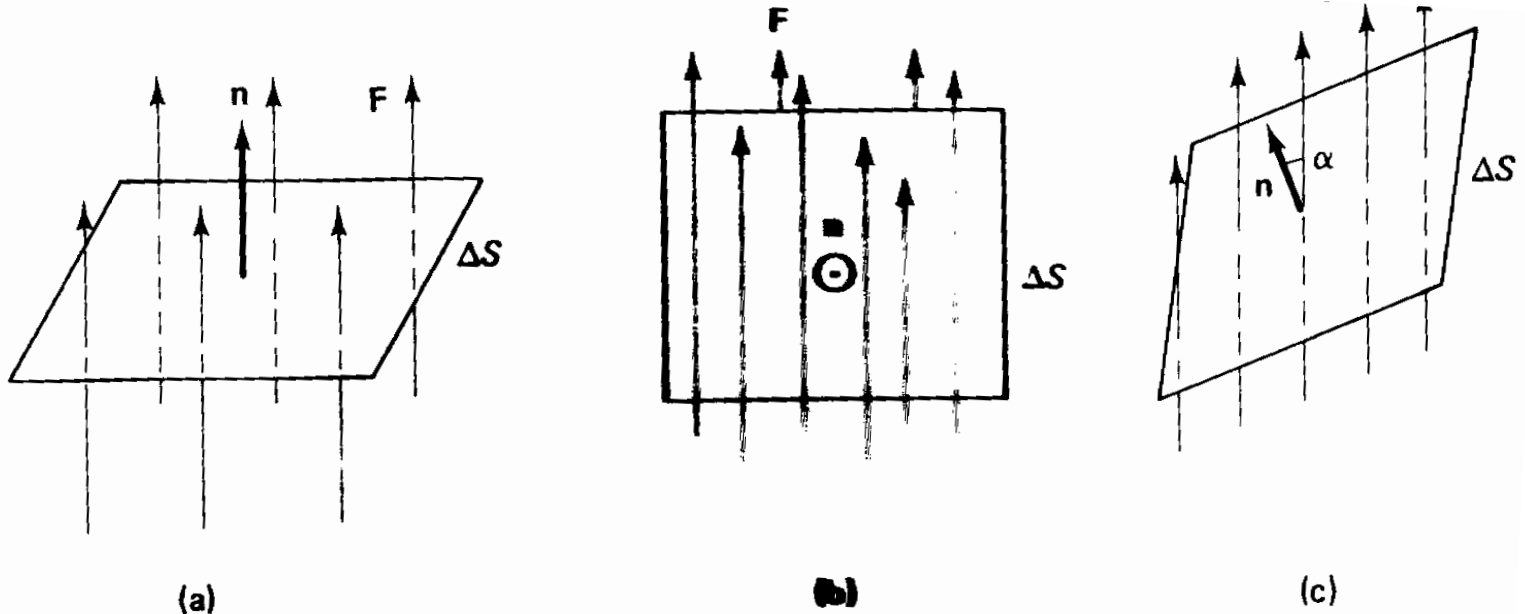


Figure 1.41 Calculation of the total vector flux crossing an element of area ΔS . (a) The flux of $\mathbf{F}$ is perpendicular to the area (i.e., parallel to unit normal $\mathbf{n}$) and the total flux crossing the area is $|\mathbf{F}|\Delta S$. (b) The flux is parallel to the area (i.e., perpendicular to unit normal $\mathbf{n}$) and the total flux crossing the area is zero. (c) The flux is making an angle α with the unit normal to the area $\mathbf{n}$. The total flux crossing the area in this case is $|\mathbf{F}| \cos \alpha \Delta S$.

Example: Flux Through Tilted Surface

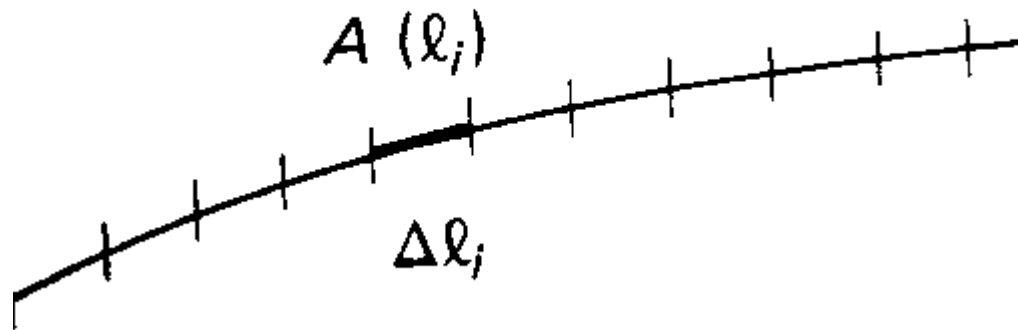
- Given: $|A| = 5$, $S = 2$, $\theta = 60^\circ$
- $\Phi = A S \cos(\theta)$
- $\Phi = 5 \times 2 \times \cos(60^\circ)$
- $\Phi = 10 \times 0.5 = 5$

Interpretation

- Maximum flux when $\theta = 0^\circ$
- Zero flux when $\theta = 90^\circ$
- Flux measures how much field passes through surface

Line Integrals

$$\int_c A(\ell) d\ell = \lim_{\substack{\Delta\ell_i \rightarrow 0 \\ N \rightarrow \infty}} \sum_{i=1}^N A(\ell_i) \Delta\ell_i$$



Example: calculating work done

$$\Delta W_i = q \underbrace{E_i \cos \alpha_i}_{\substack{\text{component} \\ \text{of force} \\ \text{along the} \\ \text{the contour}}} \Delta \ell_i$$

The total work done W in moving the charge q is then given by

$$\begin{aligned} W &= q \sum_{i=1}^N E_i \cos \alpha_i \Delta \ell_i \\ &= q \sum_{i=1}^N \mathbf{E}_i \cdot \mathbf{t} \Delta \ell_i \end{aligned}$$

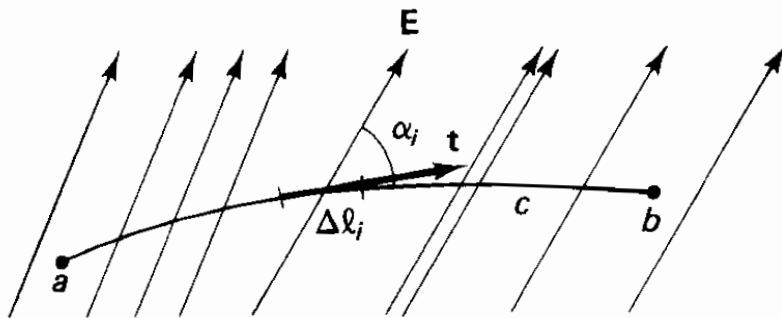


Figure 1.38 A charge q is moving along the contour c in the electric field.

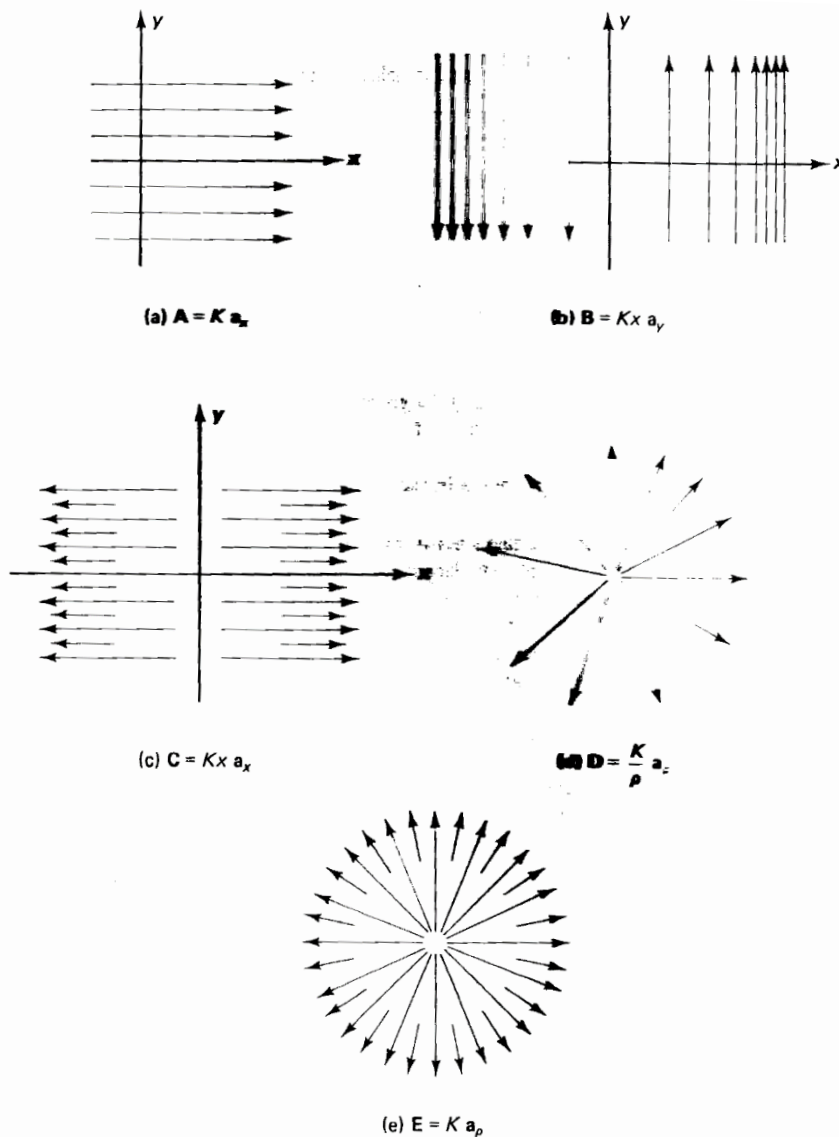


Figure 1.27 Flux representation of various vectors.

lines with the increase in ρ , as shown in Figure 1.27e, so as to maintain the density of the flux lines (i.e., number of flux lines per unit area) almost constant throughout Figure 1.27e.



Harder Example: Spherical Vectors with Angle Dependence

Given

Let

$$\mathbf{A} = 2\mathbf{a}_r - \mathbf{a}_\theta + 3\mathbf{a}_\phi$$

$$\mathbf{B} = \mathbf{a}_r + 4\mathbf{a}_\theta - 2\mathbf{a}_\phi$$

at the point

$$\theta = \frac{\pi}{3}, \quad \phi = \frac{\pi}{4}$$

We will:

1. add the vectors in spherical form
2. compute dot and cross products in spherical form
3. convert both vectors to Cartesian coordinates using the actual angles

1. Addition in spherical coordinates

Since both vectors are given at the same point, we add components directly:

$$\mathbf{A} + \mathbf{B} = (2 + 1)\mathbf{a}_r + (-1 + 4)\mathbf{a}_\theta + (3 - 2)\mathbf{a}_\phi$$

$$\boxed{\mathbf{A} + \mathbf{B} = 3\mathbf{a}_r + 3\mathbf{a}_\theta + \mathbf{a}_\phi}$$

2. Dot product

Because the spherical basis is orthonormal at a given point,

$$\mathbf{a}_r \cdot \mathbf{a}_r = 1, \quad \mathbf{a}_\theta \cdot \mathbf{a}_\theta = 1, \quad \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

and mixed products are zero.

So,

$$\mathbf{A} \cdot \mathbf{B} = 2(1) + (-1)(4) + 3(-2)$$

$$= 2 - 4 - 6 = -8$$

$$\boxed{\mathbf{A} \cdot \mathbf{B} = -8}$$

3. Cross product in spherical coordinates

Use the cyclic rule

$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi, \quad \mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r, \quad \mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$

For components $(A_r, A_\theta, A_\phi) = (2, -1, 3)$ and $(B_r, B_\theta, B_\phi) = (1, 4, -2)$,

$$\mathbf{A} \times \mathbf{B} = (A_\theta B_\phi - A_\phi B_\theta)\mathbf{a}_r + (A_\phi B_r - A_r B_\phi)\mathbf{a}_\theta + (A_r B_\theta - A_\theta B_r)\mathbf{a}_\phi$$

Substitute:

$$= [(-1)(-2) - 3(4)]\mathbf{a}_r + [3(1) - 2(-2)]\mathbf{a}_\theta + [2(4) - (-1)(1)]\mathbf{a}_\phi$$

$$= (2 - 12)\mathbf{a}_r + (3 + 4)\mathbf{a}_\theta + (8 + 1)\mathbf{a}_\phi$$

$$\mathbf{A} \times \mathbf{B} = -10\mathbf{a}_r + 7\mathbf{a}_\theta + 9\mathbf{a}_\phi$$

4. Why angle dependence matters

In spherical coordinates, the unit vectors depend on position:

$$\mathbf{a}_r = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z$$

$$\mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

So unlike Cartesian basis vectors, $\mathbf{a}_r$, $\mathbf{a}_\theta$, $\mathbf{a}_\phi$ rotate as θ , ϕ change.

5. Evaluate the basis vectors at $\theta = \pi/3$, $\phi = \pi/4$

Use

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

Then

$$\mathbf{a}_r = \frac{\sqrt{6}}{4} \mathbf{a}_x + \frac{\sqrt{6}}{4} \mathbf{a}_y + \frac{1}{2} \mathbf{a}_z$$

$$\mathbf{a}_\theta = \frac{\sqrt{2}}{4} \mathbf{a}_x + \frac{\sqrt{2}}{4} \mathbf{a}_y - \frac{\sqrt{3}}{2} \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\frac{\sqrt{2}}{2} \mathbf{a}_x + \frac{\sqrt{2}}{2} \mathbf{a}_y$$

6. Convert $\mathbf{A}$ to Cartesian coordinates

$$\mathbf{A} = 2\mathbf{a}_r - \mathbf{a}_\theta + 3\mathbf{a}_\phi$$

Substitute the basis vectors:

$$\mathbf{A} = 2 \left(\frac{\sqrt{6}}{4} \mathbf{a}_x + \frac{\sqrt{6}}{4} \mathbf{a}_y + \frac{1}{2} \mathbf{a}_z \right) - \left(\frac{\sqrt{2}}{4} \mathbf{a}_x + \frac{\sqrt{2}}{4} \mathbf{a}_y - \frac{\sqrt{3}}{2} \mathbf{a}_z \right) + 3 \left(-\frac{\sqrt{2}}{2} \mathbf{a}_x + \frac{\sqrt{2}}{2} \mathbf{a}_y \right)$$

Now combine components:

x -component:

$$\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{4} - \frac{3\sqrt{2}}{2} = \frac{2\sqrt{6} - 7\sqrt{2}}{4}$$

y -component:

$$\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{4} + \frac{3\sqrt{2}}{2} = \frac{2\sqrt{6} + 5\sqrt{2}}{4}$$

z -component:

$$1 + \frac{\sqrt{3}}{2}$$

So,

$$\mathbf{A} = \frac{2\sqrt{6} - 7\sqrt{2}}{4} \mathbf{a}_x + \frac{2\sqrt{6} + 5\sqrt{2}}{4} \mathbf{a}_y + \left(1 + \frac{\sqrt{3}}{2} \right) \mathbf{a}_z$$

7. Convert $\mathbf{B}$ to Cartesian coordinates

$$\mathbf{B} = \mathbf{a}_r + 4\mathbf{a}_\theta - 2\mathbf{a}_\phi$$

Substitute:

$$\mathbf{B} = \left(\frac{\sqrt{6}}{4}\mathbf{a}_x + \frac{\sqrt{6}}{4}\mathbf{a}_y + \frac{1}{2}\mathbf{a}_z \right) + 4 \left(\frac{\sqrt{2}}{4}\mathbf{a}_x + \frac{\sqrt{2}}{4}\mathbf{a}_y - \frac{\sqrt{3}}{2}\mathbf{a}_z \right) - 2 \left(-\frac{\sqrt{2}}{2}\mathbf{a}_x + \frac{\sqrt{2}}{2}\mathbf{a}_y \right)$$

Combine components:

x -component:

$$\frac{\sqrt{6}}{4} + \sqrt{2} + \sqrt{2} = \frac{\sqrt{6}}{4} + 2\sqrt{2}$$

y -component:

$$\frac{\sqrt{6}}{4} + \sqrt{2} - \sqrt{2} = \frac{\sqrt{6}}{4}$$

z -component:

$$\frac{1}{2} - 2\sqrt{3}$$

So,

$$\mathbf{B} = \left(\frac{\sqrt{6}}{4} + 2\sqrt{2} \right) \mathbf{a}_x + \frac{\sqrt{6}}{4} \mathbf{a}_y + \left(\frac{1}{2} - 2\sqrt{3} \right) \mathbf{a}_z$$